

Q1. Asynchronous Serial Data Format

1. Figure 1 shows the logical waveform of one asynchronous serial data frame.

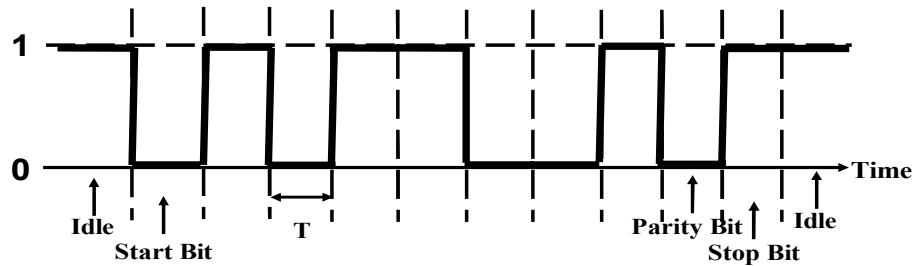


Figure 1

Answer the following questions with reference to Figure 1:

- Specify the 7-bit ASCII character that is being transmitted (LSB is transmitted first).
- Assuming there are no errors in the transmitted waveform, state whether even or odd parity is being used.
- Assuming that 2 of the bits received in this character are erroneous; would the receiver be able to detect this? Could you extend this example to determine the limitations of parity checking?
- If 7-bit ASCII characters are transmitted continuously in the format shown (with no idle periods between data frames) and at a baud rate of 9600, calculate the following:
 - The value of T in the Figure 1.
 - The data transfer rate of this serial interface in characters per second.
- Re-compute question 1(d)(ii) if we now transmit without using any parity bit. Discuss what has been sacrificed for this gain in transfer rate.

Q2. Data Transfer Techniques (Polled I/O and Interrupt-Driven I/O)

- 2.1 A serial communications I/O module is configured to receive data using the asynchronous data format with 7-data bits, 1 parity bit and 1 stop bit. The module will interrupt the CPU every time it receives a character. The CPU will then execute the Interrupt Service Routine (ISR) to read the character received. Assuming that each interrupt takes a minimum of $100\mu\text{s}$ (from the time the interrupt request is generated) to service, what is the maximum baud that can be supported by this serial communication I/O module?

2.2 Assume that a keypad of a microprocessor system is pressed 90 times within an 8-hour interval.

- Suppose the microprocessor scans the keypad every 100ms using the polled I/O technique. How many times will the keypad be scanned in an 8-hour period?
- By what fraction would the number of keypad accesses by the microprocessor be reduced if interrupt-driven I/O were used?

LS \ MS	0	1	2	3	4	5	6	7
0	NUL	DLE	SP	0	@	P	'	p
1	SOH	DC1	!	1	A	Q	a	q
2	STX	DC2	"	2	B	R	b	r
3	ETX	DC3	#	3	C	S	c	s
4	EOT	DC4	\$	4	D	T	d	t
5	ENQ	NAK	%	5	E	U	e	u
6	ACK	SYN	&	6	F	V	f	v
7	BEL	ETB	'	7	G	W	g	w
8	BS	CAN	(	8	H	X	h	x
9	HT	EM	)	9	I	Y	i	y
A	LF	SUB	*	:	J	Z	j	z
B	VT	ESC	+	;	K	[	k	{
C	FF	FS	,	<	L	\	l	
D	CR	GS	-	=	M	]	m	}
E	SO	RS	.	>	N	^	n	
F	SI	US	/	?	O	_	o	DEL

The ASCII Table